

JEN-WEI (Wayne) KUO 郭人瑋

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ABOUT ME

M.S. in Computer and Communication Engineering at NCKU, focused on DRL-based wireless optimization and **deployable AI systems**. Built **Jetson-targeted edge-AI/GenAI applications** across audio, real-time vision, gait authentication, and ControlNet diffusion.

Proven ML competitiveness: Kaggle Silver Top 0.66% (15/2,240) + multiple top-10% rankings.

EDUCATION

National Cheng Kung University

09/2024 – 06/2026

M.S. in Computer and Communication Engineering

- Advisor: Prof. Wenson Chang (張志文) — Intelligent Communication Lab
- **Research:** Voronoi-based energy-aware user partitioning and DRL resource allocation for dual-UAV RIS-assisted SWIPT networks, jointly optimizing time switching, power allocation, and RIS phase shifts to improve energy-harvesting efficiency.

National Cheng Kung University

09/2020 – 06/2024

B.S. in Systems and Naval Mechatronic Engineering

- Capstone advisor: Prof. Chien-Hsing Lee (李建興), Lab617 — Intelligent Unmanned Vessel project (see below).

COMPETITIONS & AWARDS

- Kaggle NeurIPS 2025 - Open Polymer Prediction - Silver Medal, Top 0.66% (15/2,240)
- CVPR 2026 CV4CHL Workshop - The First AI for Children Challenge - Gold Award / 2nd Place (Team CCE)
- Kaggle Playground S5E8 - Binary Classification with a Bank Dataset - Top 6.7% (227/3,377)
- Kaggle Playground S6E3 - Predict Customer Churn - Top 6.9% (286/4,142)
- Taiwan Future Accident Classification Challenge (TAISC @ IEEE AVSS 2025) - Top 8.9% (12/134)
- AI CUP 2025 Spring - Table Tennis Smart Racket Analysis - Top 7.4% (47/633)
- SinoPac AI GO Stock Prediction Challenge (Financial Forecasting) - Top 13.9% (121/868)
- Kaggle Community Competition (MLB) - Top 13% (19/142)

PUBLICATIONS

KUO Jen Wei, Yi-Ting Ku.

Clinically-Grounded Multi-View Features and Cross-Track Rule Coupling for Children Gait Visual Analysis.

Accepted as a non-proceedings workshop submission, CVPR 2026 Workshop on Computer Vision for Children (CV4CHL), 2026.

SELECTED PROJECTS

AIDSP Edge AI & GenAI Deployment Suite 03/2026 – 06/2026

Stack: Jetson Xavier NX · FastAPI/Gradio · Docker · ONNX Runtime/TensorRT FP16 · YOLOv8-pose · Stable Diffusion/ControlNet · Gemma

- **Edge AI pipeline:** Built Jetson-targeted audio + LLM and real-time vision demos: 20 s audio capture -> log-mel -> ONNX genre inference -> Gemma lyrics, plus neural style transfer via ONNX/TensorRT FP16 with 15 FPS webcam streaming.
- **Gait + diffusion demos:** Built YOLOv8-pose gait authentication/tailgating alerts and a public ControlNet Canny/Sketch UI with scheduler, seed, CFG, weight controls, and ablation logging.

Intelligent Unmanned Vessel — Capstone Design Practicum

03/2023 – 01/2024

Role: Sound-source recognition · electromechanical integration | Stack: Raspberry Pi · Arduino Mega / UNO R4 Wi-Fi · Python · C++

- **Hull engineering:** Designed vessel in **Rhino + Orca 3D** (hydrostatics, FEA, resistance analysis) and built a balsa-wood hull with epoxy waterproofing and propulsion installation — drag values matched Holtrop theoretical estimates.
- **Autonomous heading control:** Implemented **PID + bang-bang closed-loop control** (MPU-9250 gyroscope + servo rudder) on separate MCUs to eliminate processing-delay conflicts.
- **Acoustic navigation:** Built a **real-time sound-source classification pipeline** (band-pass 500–3500 Hz → Log-MMSE denoising → cross-correlation matching) achieving accurate rising / flat / falling tone **recognition under motor noise**.
- **Computer vision:** Trained **YOLOv8** (3,000 frames, Tesla T4 GPU, precision 0.998 / AP@0.5 0.995) with **adaptive histogram equalization + Gaussian blur augmentation** for **underwater LED digit recognition**; resolved 2.5–3 s Pi↔Arduino USB latency by switching to **direct Pi servo control**.
- **Additional subsystems:** HC-SR04P ultrasonic obstacle avoidance and photoresistor light-source tracking; fully integrated into a sequential multi-modal pipeline.

Brain Tumor MRI Segmentation — Attention U-Net + Elastic Augmentation

09/2025 – 11/2025

- Implemented custom Attention U-Net with elastic deformation augmentation for LGG MRI segmentation.
- Achieved validation Dice 0.9070 / Dice 0.9116, surpassing the Kaggle U-Net baseline (Dice approx. 0.896) with improved recall and boundary fit.

Unified-OneHead Multi-Task Learning Model

04/2025 – 06/2025

- Shared-encoder + single-head architecture covering classification, detection, and segmentation; Replay Buffer mitigates catastrophic forgetting - performance degradation held within 5% across all tasks.

DynamicConv — Input-Conditioned Lightweight Convolution Module

03/2025 – 06/2025

- **MLP-generated dynamic kernels** supporting multi-channel input combinations (RGB / RG / R); outperforms static baselines in flexibility and inference efficiency.

TinyViT: Transformer-based Lightweight Vision Model

03/2025 – 05/2025

- 1.1M parameters / 5M FLOPs - matches ResNet-34 (90% accuracy); ablation on FFN size and patch resolution validated design efficiency.

AI Mock-Interview Platform with GPT-4o & RAG

04/2025 – 06/2025

- Streamlit + LangChain + OpenAI API; vector-DB-backed RAG for resume-aware Q&A; Dockerized for one-command deployment; fine-tuned feedback loop adapts to company culture profiles.

Traffic Accident Prediction — CLIP Semantic Features + XGBoost

04/2024 – 07/2024

- Top 9.2% (12/130) at IEEE AVSS TAISC 2025; CLIP-derived semantic similarity features combined with gradient-boosted trees for accident-risk scoring from image data.

SKILLS

Programming: Python · C/C++ · MATLAB · FORTRAN

ML / CV / GenAI: PyTorch · TensorFlow · scikit-learn · OpenCV · YOLOv8/YOLOv8-pose · CLIP · Stable Diffusion/ControlNet · Diffusers

Deployment / MLOps: Docker/Compose · FastAPI · Gradio · Streamlit · ONNX Runtime · TensorRT FP16 · LangChain · OpenAI API · RAG · Git/Linux

RL: DRL-based optimization / custom environment / simulator design

Wireless / DSP: Physical-layer communications · RIS · UAV communications · Signal processing

Embedded: Raspberry Pi · STM32 · Arduino (AVR / ARM) · NVIDIA Jetson Xavier / Xavier NX

Language: Mandarin (native) · English — TOEIC 720

Links: GitHub | LinkedIn | Website